Youngjae Choi

Contact Information

Nationality:	Dual Citizenship: Republic of Korea and United States
Affiliation:	Department of Digital Media, Soongsil University, Seoul, Korea
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Education

M.S. program in Department of Digital Media *Mar. 2024 - Present*
Soongsil University, Seoul, Korea
Advisor: Prof. Heewon Kim

B.S. in Department of Statistics and Actuarial Science *Mar. 2015 - Feb. 2022*
Soongsil University, Seoul, Korea

Projects

- **AI-powered XR Smart Space Solution Development** *SME R&D Support Program*
  Participated in a national R&D project aimed at reconstructing interactive virtual spaces in AR by leveraging 3D Gaussian Splatting (3DGS). Developed a system that automatically estimates box colliders from reconstructed geometry, enabling real-time interaction within Unity-based XR environments.
- **VQA-based Annotation Tool Design for Autonomous Driving**
  Designed and developed a VQA-based annotation framework for autonomous driving datasets, supporting efficient annotation, validation, and quality control across 2D object detection, 3D object detection, and HD map tasks. Focused on reducing manual annotation effort through vision-language reasoning, automated consistency checking, and intelligent interaction with large-scale sensor data.
- **VLM-based Data Curation and Driving Scenario Mining**
  Developed VLM-based data curation and scenario mining pipelines for large-scale autonomous driving datasets. Applied vision-language models to understand, categorize, and retrieve complex driving scenarios, enabling automated filtering of high-value, rare, and safety-critical samples for training and evaluation of end-to-end driving models.
- **World Model-based Data Augmentation and Driving Simulation**
  Researched World Models for synthetic data generation, data augmentation, and simulation in end-to-end autonomous driving, including approaches inspired by large-scale driving simulation frameworks such as AlpaSim. Explored controllable generation of diverse driving scenarios, novel-view and trajectory-based simulation, and synthetic data generation to improve scenario coverage and reduce dependence on real-world data collection.

Experience

Research Intern *Jun. 2023 - Feb. 2024*
Reality Lab, Soongsil University, Seoul, Korea *Mar. 2024 - Feb. 2026*

Teaching experience

Teaching Assistant for *Deep learning* at Soongsil University (spring 2024)
Teaching Assistant for *Machine learning* at Soongsil University (Spring 2025)

Working experience

Motifdrive(Autonomous Vehicle Start-up) *Jan. 2026 -*
Researched World Models for synthetic data generation, simulation, and data augmentation in end-to-end autonomous driving, with a focus on generating diverse driving scenarios and improving data coverage for rare and challenging cases. Developed VLM-based frameworks for large-scale data management, automated data curation, scenario understanding, and scenario mining to efficiently identify high-value training and evaluation samples from autonomous driving datasets.

Worked on simulation and data augmentation projects leveraging World Models to generate synthetic driving data and explore novel-view and trajectory-based scenario generation for end-to-end driving systems. Investigated methods to bridge the domain gap between reconstructed or simulated scenes and real-world sensor data, with the goal of improving the usability of synthetic data for model training.

Developed and improved data curation and annotation workflows using VLMs and automated analysis pipelines, including tools for dataset filtering, quality control, annotation validation, and scenario categorization. Contributed to the design and improvement of annotation tools for 2D/3D object detection and HD map data, focusing on annotation efficiency, data quality, workflow reliability, and large-scale dataset management.

Also contributed to 3D object detection research using multi-modal sensor data, including LiDAR, camera, and other vehicle sensor modalities, with an emphasis on sensor fusion, domain adaptation, pseudo-label generation, and improving robustness on large-scale autonomous driving datasets.

Research interests

Computer vision, Deep learning, Motion Generation, Image Generation, Embodied AI

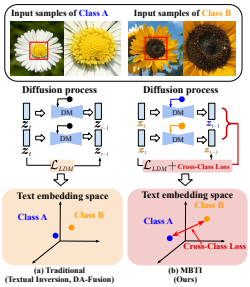
Research Highlight

Conducting research on multimodal generative models with a focus on generating fine-grained images and expressive human motions from text inputs. This includes developing methods that reflect detailed semantics and emotional context in the generation process.

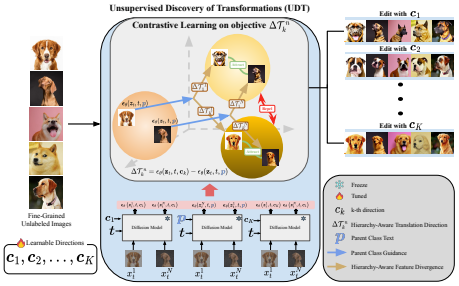
Skills

Python, PyTorch, MATLAB,Object Detection & Segmentation(MMCV), 3DGS, LaTeX

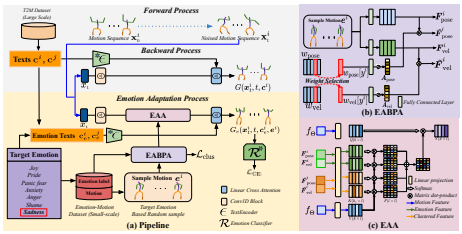
Publications



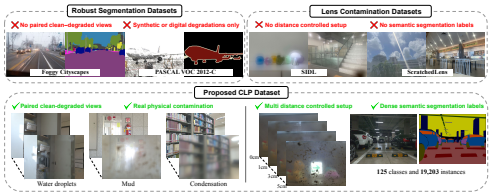
- **MBTI**: Metric-Based Textual Inversion for Fine-Grained Image Generation
  ByungKwan Chae*, **Youngjae Choi***, and Heewon Kim
  WACV 2026



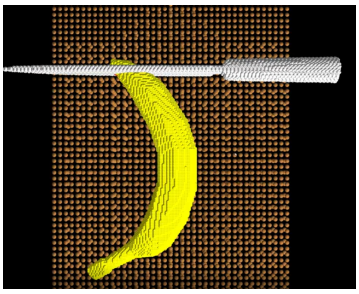
- **UDT**: Unsupervised Discovery of Transformations between Fine-Grained Classes in Diffusion Models
  Youngjae Choi*, Hyunsuh Koh*, Byungkwan Chae*, Hojae Jeong*, and Heewon Kim
  BMVC 2025



- **FEEL**: Focus on Emotion-Enriched Motion Using Unsupervised Learning for Text-to-Motion Generation
  Youngjae Choi*,Hyunsuh Koh, and Heewon Kim
  CVPRW 2026



- CLP: A Real-World Dataset of Contaminated Lens Protectors for Robust Semantic Segmentation
  Sungyong Park*, Sooyoung Choi*, Hyunseo Koh, **Youngjae Choi**, and Heewon Kim
  CVPR 2026



- CulinaryCut: A Physics-Grounded Cutting Benchmark for Vision-Language-Action Models
  Hyunseo Koh*, **Youngjae Choi***, Chang-Yong Song*, Misa Viveiros, David Hyde, Heewon Kim
  ECCV 2026

References

Advisor — Heewon Kim
Professor
Soongsil University
hwkim(at)ssu.ac.kr
https://sites.google.com/view/kimheewon/